SOLUTION

SECTION–I

1. Ans. (C)

$\vec{x}, \vec{y}$ & $\vec{x}\times\vec{y}$ are linearly independent

$a\alpha^2 + b\alpha + c = 0$

$a\beta^2 + b\beta + c = 0$

$a\gamma^2 + b\gamma + c = 0$

$\Rightarrow$ quadratic $ax^2 + bx + c = 0$ has three roots α,β and γ

$\Rightarrow ax^2 + bx + c = 0$ is an identity

$\Rightarrow a = b = c = 0$

2. Ans. (C)

$\hat{a}\times\hat{b} + \hat{a} = \vec{c}$ ...(i)

$\hat{a}\times\hat{b} = \vec{c} - \hat{a}$

Dot with $\vec{c} \Rightarrow v = \left[\hat{a}\ \hat{b}\ \vec{c}\right] = |\vec{c}|^2 - \hat{a}.\hat{c}$ ...(ii)

for (i) : dot with $\hat{b}$

$\Rightarrow 0 + \hat{a}.\hat{b} = \hat{b}.\vec{c} = 0 \Rightarrow \hat{a}.\hat{b} = 0$

for (i) dot with $\hat{a} \Rightarrow 0 + 1 = \hat{a}.\vec{c}$

(ii) $\Rightarrow \left[\hat{a}\ \ \hat{b}\ \ \vec{c}\right] = |\vec{c}|^2 - 1$

now $\hat{a} = \vec{c} - \hat{a}\times\hat{b}$

$|\hat{a}|^2 = \left|\vec{c} - \left(\vec{a}\times\hat{b}\right)\right|^2$

$|\hat{a}|^2 = \left|\vec{c} - \left(\vec{a}\times\hat{b}\right)\right|^2$

$\Rightarrow 1 = |c|^2 + \left|\hat{a}\times\hat{b}\right|^2 - 2\left[\hat{a}\ \ \hat{b}\ \ \vec{c}\right]$

$\Rightarrow 1 = \left[\hat{a}\ \ \hat{b}\ \ \vec{c}\right] + 1 + 1 - 2\left[\hat{a}\ \ \hat{b}\ \ \vec{c}\right]$

$\Rightarrow \left[\hat{a}\ \ \hat{b}\ \ \vec{c}\right] = 1$

3. Ans. (B)

$\vec{d}.\vec{c} = \lambda\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right] + \mu.0 + v.0$

$\vec{d}.\vec{a} = \mu\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]$

$\vec{d}.\vec{b} = v\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]$

$\vec{d}.\left(\vec{a}\ \ \vec{b}\ \ \vec{c}\right) = \left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right](\lambda + \mu + v)$

$\lambda + \mu + v = 64$

4. Ans. (A)

Let $|\vec{U}| = 2\cos\alpha\hat{i} + 2\sin\alpha\hat{j}$

$$\left([\vec{U}\ \vec{V}\ \vec{W}]\right)^2 = \begin{vmatrix} 2\cos\alpha & 2\sin\alpha & 0 \\ 2 & 1 & -1 \\ 1 & 0 & 3 \end{vmatrix}^2$$

$= |6\cos\alpha - 14\sin\alpha|^2$

Maximum value = 36 + 196 = 232

5. Ans. (B)

Let $A(\vec{a}), B(\vec{b})$ and $C(\vec{c})$

$\Rightarrow G_1\left(\frac{\vec{a}+\vec{b}+\vec{c}}{3}\right)$ and $G_2\left(\frac{\vec{a}+\vec{c}}{3}\right)$

$\overrightarrow{OG_1}\cdot\overrightarrow{BG_2} = 0 \Rightarrow \frac{\vec{a}+\vec{b}+\vec{c}}{3}\cdot\frac{a+\vec{c}-3\vec{b}}{3} = 0$

$\Rightarrow\ a^2 + c^2 - 3b^2 + 2\vec{a}\cdot\vec{c} - 2\vec{b}\cdot\vec{c} - 2\vec{a}\cdot\vec{b} = 0$

Now, $|\vec{c}-\vec{a}|^2 = b^2, |\vec{c}-\vec{b}|^2 = a^2$ and $|\vec{a}-\vec{b}|^2 = c^2$

{$\because$ AC = b, BC = a and AB = c}

$\Rightarrow 2\vec{a}\cdot\vec{c} = a^2 + c^2 - b^2, 2\vec{b}\cdot\vec{c}$

$= b^2 + c^2 - a^2, 2\vec{a}\cdot\vec{b} = a^2 + b^2 - c^2$

Putting in the above result, we get

$2a^2 + 2c^2 - 6b^2 = 0 \quad \Rightarrow \quad \frac{a^2+c^2}{b^2} = 3$

6. Ans. (B)

Required plane

$(1 + 2\lambda)x + (2 - 3\lambda)y + (3 + 7\lambda)z + 5 + \lambda = 0$

it is parallel to given line

$\Rightarrow$

$(1+2\lambda)\hat{i} + (2-3\lambda)\hat{j} + (3+7\lambda)\hat{k}.(8\hat{i} - 7\hat{j} - 4\hat{k}) = 0$

$\Rightarrow 8(1+2\lambda) - 7(2-3\lambda) - 4(3+7\lambda) = 0$

$\Rightarrow \lambda = 2$

$x\text{ intercept} = \frac{-5-\lambda}{1+2\lambda} = -\frac{7}{5}$

7. Ans. (B)

Let the plane be $a(x-1) + b(y-0) + c(z-0) = 0$

$\Rightarrow ax + by + cz - a = 0$

Pass through (0,1,0) $\Rightarrow a.0 + b.1 + c.0 - a = 0$

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$\Rightarrow a = b$ ...(i)

Normal to plane : $a\hat{i} + b\hat{j} + c\hat{k}$

Makes angle $\frac{\pi}{4}$ with $\hat{i} + \hat{j}$

$\therefore \frac{a+b}{\sqrt{a^2+b^2+c^2}.\sqrt{2}} = \frac{1}{\sqrt{2}}$

$\Rightarrow (a+b)^2 = a^2 + b^2 + c^2$ ...(ii)

By (i) and (ii) : $4a^2 = 2a^2 + c^2 \Rightarrow 2a^2 = c^2$

$\Rightarrow \sqrt{2}a = c; -\sqrt{2}a = c$

$\therefore a : b : c = 1 : 1 : \sqrt{2}$

8. Ans. (B)

$\frac{x-1}{3} = \frac{y-2}{1} = \frac{z-3}{2} = t$

$\frac{x-3}{1} = \frac{y-1}{2} = \frac{z-2}{3} = \mu$

for point of intersection

$3t + 1 = \mu + 3$ ...(1)

$t + 2 = 2\mu + 1$ ...(2)

$2t + 3 = 3\mu + 2$

from (1) and (2)

$5 = 5\mu$

$\Rightarrow \mu = 1$

$\therefore t = 1$

$\therefore$ Point of intersection P(4,3,5)

Now, Plane passing through P and at greatest distance from (0,0,0) will have its normal vector along $\overrightarrow{OP}$.

$\therefore$ Required equaiton : $4(x-4) + 3(y-3) + 5(z-5) = 0$

$4x + 3y + 5z = 50$

option (B)

9. Ans. (A)

Since the line $\frac{x-1}{2} = \frac{y}{-1} = \frac{z+2}{1}$ is parallel to the plane $2x + y - 3z = 4$

($\because 2 \times 2 - 1 - 3 = 4 - 4 = 0$).

so the direction cosines of the projected line remains the same as the given line

so dc's = $\frac{2}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}$

10. Ans. (A)

Let $\vec{a} = x\hat{i} + y\hat{j} + z\hat{k}$

$\therefore x^2 + y^2 + z^2 = 1$ ...(1)

also $\vec{a} \times (\hat{i} + \hat{j} + \hat{k}) = \hat{i} - \hat{k}$

$\Rightarrow \hat{i}(y-z) - \hat{j}(x-z) + \hat{k}(x-y) = \hat{i} - \hat{k}$

comparing, we get

$z = x; y = 1 + x$

$\therefore$ By (1): $x = -\frac{2}{3}; y = \frac{1}{3}; z = -\frac{2}{3}$ or $x = z = 0, y = 1$

$\therefore \hat{a} = -\frac{2}{3}\hat{i} + \frac{1}{3}\hat{j} - \frac{2}{3}\hat{k}$

$\therefore \hat{a}.\hat{b} = -\frac{2}{3} + \frac{1}{3} - \frac{2}{3} = -\frac{3}{3} = -1$

11. Ans. (A,C)

$\begin{vmatrix} 1 & 6 & 3 \\ 3 & 2 & 1 \\ \alpha+1 & \beta-1 & 1 \end{vmatrix} = 0 \Rightarrow \beta = 3$

$|\vec{c}| = \sqrt{6} \Rightarrow \alpha = 0$ or -2

12. Ans. (A,C,D)

$a_1 + a_2\cos 2x + a_3 + \sin^2 x = 0 \ \forall \ x \in R$

$\Rightarrow (a_1 + a_2) + (a_3 - 2a_2)\sin^2 x = 0 \ \forall \ x \in R$

$\therefore a_1 = -a_2 = -\frac{a_3}{2} = \lambda$

$\therefore \vec{a} = \lambda\hat{i} - \lambda\hat{j} - 2\lambda\hat{k}$

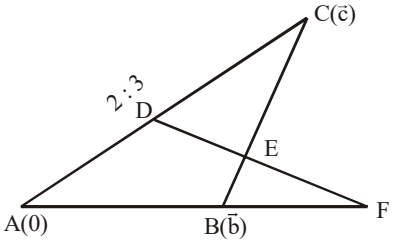
if $|\vec{a}| = \sqrt{6} \Rightarrow \lambda = \pm 1$

$\Rightarrow$ ordered triplet $(1, -1, -2)$ or $(-1, 1, 2)$

if $2a_1 + 3a_2 + 6a_3 = 26 \Rightarrow \lambda = -2$

$\Rightarrow |\vec{a}| = 2\sqrt{6}$

13. Ans. (A,C)



$D\left(\frac{2\vec{c}}{5}\right)$ & $E\left(\frac{\vec{c} + 2\vec{b}}{3}\right)$

Let DE : EF = λ : 1 & AB : BF = μ : 1

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$$\frac{\vec{c}+2\vec{b}}{3}=\frac{\lambda\vec{f}+\frac{2\vec{c}}{5}}{\lambda+1} \quad \text{.....(i)}$$

& $\vec{b}=\frac{\mu\vec{f}}{\mu+1}\Rightarrow\vec{f}=\frac{\mu+1}{\mu}\vec{b}$

$$\frac{\vec{c}+2\vec{b}}{3}=\frac{\frac{\lambda(\mu+1)\vec{b}}{\mu}+\frac{2\vec{c}}{5}}{\lambda+1}$$

$$\frac{1}{3}=\frac{2}{5(\lambda+1)} \quad \& \ 5\lambda+5=6$$

$$\lambda=\frac{1}{5} \quad \& \quad \mu=\frac{1}{3}$$

14. Ans. (A,B,C,D)

$$\cos\alpha=\frac{\vec{u}.\vec{v}}{|\vec{u}|.|\vec{v}|}=\frac{2x^2-3x+1}{\sqrt{x^2+9+1}.\sqrt{4x^2+x^2+1}}>0$$

$\Rightarrow \quad (2x-1)(x-1)>0$

$\Rightarrow \quad x\in\left(-\infty,\frac{1}{2}\right)\cup(1,\infty)$ .....(1)

$$\cos\beta=\frac{\vec{v}.\hat{j}}{|\vec{v}|}=\frac{x}{\sqrt{4x^2+x^2+1}}<0$$

$\Rightarrow \quad x<0$ .....(2)

from (1) & (2), $x<0 \Rightarrow x^2\in R_0$

15. Ans. (A,B,C)

$\bar{a}\times\bar{b}+\bar{b}=\bar{c}\times\bar{a}+\bar{b}\times\bar{c}$ .....(i)

taking dot product with $\bar{a}$ and $\bar{b}$ respectively

$\bar{a}.\bar{b}=\left[\bar{a}\,\bar{b}\,\bar{c}\right]$

& $|\bar{b}|^2=\left[\bar{a}\,\bar{b}\,\bar{c}\right]$

Let $|\bar{a}|=|\bar{b}|=k$

$\Rightarrow \left[\bar{a}\,\bar{b}\,\bar{c}\right]=k^2\cos\theta$

& $\left[\bar{a}\,\bar{b}\,\bar{c}\right]=k^2$

$\Rightarrow \cos\theta=1$ where $\theta=\bar{a}\wedge\bar{b}$

$\Rightarrow \bar{a}=\bar{b}$

putting in (i)

$\bar{a}=\bar{c}\times\bar{a}+\bar{a}\times\bar{c}$

$\Rightarrow \bar{a}=\bar{0}$

$\Rightarrow \bar{b}=\bar{0}$

and $\bar{c}$ can be any general vector

16. Ans. (A,B,C)

Equation of P_1

$2(x-1)-2(y+1)+2(z-1)=0$

$\Rightarrow x-y+z-3=0$ 1

Equation of P_2 is

$(-2)(x-1)+3(y-2)+2(z-1)=0$

$\Rightarrow 2x-3y-2z+6=0$ 2

Vector along line of intersection of plane P_1 and P_2 is

$$\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&-1&1\\2&-3&-2\end{vmatrix}=5\hat{i}+4\hat{j}-\hat{k}$$

dr of line is 5, 4, –1

in (1) and (2) put $z=0$, we get

$x=15, y=12$

Equation of line of intersection of P_1 and P_2 is

$$\frac{x-15}{5}=\frac{y-12}{4}=\frac{z-0}{-1}$$

$P_3: P_1+\lambda P_2=0$

$(x-y+z-3)+\lambda(2x-3y-2z+6)=0$

Passes through (1, 0, –1)

$\Rightarrow 16x-19y+4z=12$

$\therefore$ length of $\perp$ from (0, 0, 0) to P_3 is $=\frac{12}{\sqrt{633}}$

Paragraph for Question 17 and 18

17. Ans. (C)

Equation of line is $(\vec{r}-\vec{a})\times\vec{b}=\vec{0}\Rightarrow\vec{r}=\vec{a}+t\vec{b}$

$\Rightarrow \vec{r}=\hat{i}+2\hat{j}-3\hat{k}+t\left(2\hat{i}-\hat{j}+\hat{k}\right)$

Let M be the foot of $\perp$ from P(0,1,2) to the line

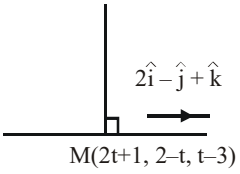
$\therefore \ \overline{PM}=(2t+1)\hat{i}+(1-t)\hat{j}+(t-5)\hat{k}$

$\because \ \overline{PM}.\left(2\hat{i}-\hat{j}+\hat{k}\right)=0$

$2(2t+1)-1.(1-t)+1.(t-5)=0$

$4t+2-1+t+t-5=0$

$\Rightarrow 6t=4\Rightarrow t=\frac{2}{3}$



$\therefore \ M=\left(\frac{7}{3},\frac{4}{3},-\frac{7}{3}\right)$ $\therefore$ Shortest distance = PM

$$=\sqrt{\frac{49}{9}+\frac{1}{9}+\frac{169}{9}}=\frac{\sqrt{219}}{3}$$

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18. Ans. (B)

Let $\vec{R} = x\hat{i} + y\hat{j}$ then curve 'C' is circle

$x^2 + y^2 - 2x + 2y + 1 = 0$ whose centre = (1,–1)

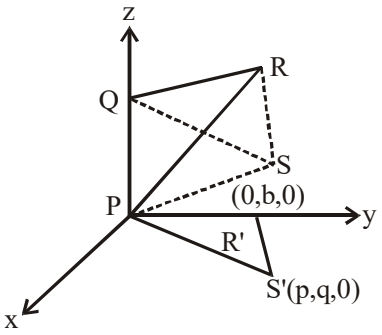
Now proceeding as above problem we can find minimum distance of line 'L' from the point (1,–1,0). Min. distance $= 2\sqrt{3}$.

Paragraph for Question 19 & 20

P(0, 0, 0); Q(0, 0, 4)

R(0, b, c); S(p, q, r)

P is same as P'



$\text{Ar}(\Delta PRS') = \frac{1}{2}bp = 15$

bp = 30

$\text{Vol}(PQRS) = \frac{1}{3} \times (\text{ArPQR}) \times \text{height}$

$= \frac{1}{2} \times \left(\frac{1}{2} \times 4 \times b\right) \times p$

$= \frac{1}{3} \times \frac{1}{2} \times 4 \times 30 = 20$ cu. units.

if P = 5, b = 6

19. Ans. (B)

20. Ans. (B)

Paragraph for Questions 21 and 22

21. Ans. (C)

Let P(cosθ, sinθ, 0) be a point on the circle then its distance d from the

$$\text{plane} = d = \left|\frac{3 - \cos\theta - \sin\theta}{\sqrt{3}}\right| = \left|\frac{3 - \sqrt{2}\sin\left(\theta + \frac{\pi}{4}\right)}{\sqrt{3}}\right|$$

for maxima, $\theta = \frac{-3\pi}{4}$

and for minima, $\theta = \frac{\pi}{4}$

$\Rightarrow$ $A\left(-\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, 0\right)$ and $B\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)$

$\Rightarrow$ $|x_1 + y_1 - x_2 - y_2| = 2\sqrt{2}$

22. Ans. (D)

Plane P is x + y +z = 3

get the image of P(cosθ, sinθ, 0) in x + y + z = 3 which comes out as

$$P'\left(2 + \frac{\cos\theta - 2\sin\theta}{3}, 2 + \frac{\sin\theta - 2\cos\theta}{3}, 2 - \frac{2}{3}(\sin\theta + \cos\theta)\right)$$

i.e. $x = 2 + \frac{\cos\theta - 2\sin\theta}{3}$

$y = 2 + \frac{\sin\theta - 2\cos\theta}{3}$

and $z = 2 - \frac{2}{3}(\sin\theta + \cos\theta)$

$\Rightarrow$ eliminating θ, we get 2x + 2y – z = 6

23. Ans. (A)

(P) $\vec{x} \times \vec{b} = \vec{c} \times \vec{b}$

$\Rightarrow$ $\vec{x} - \vec{c} = t\vec{b}$, t is scalar

$\vec{x} = \vec{c} + t\vec{b}$

$\vec{x} = \vec{c} + t\vec{b}$

$\vec{x}.\vec{a} = 0 \Rightarrow t = -\left(\frac{\vec{c}.\vec{a}}{\vec{b}.\vec{a}}\right) = -\left(\frac{8+7}{2+1}\right) = 5$

$\vec{x} = \vec{c} - 5\vec{b}$

$\vec{x} - \vec{c} = -5\vec{b} = -5\left(\hat{i} + \hat{j} + \hat{u}\right)$

(Q) $\vec{x} \times \vec{a} + \vec{x} \times \vec{b} = 0$

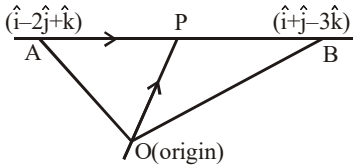
$\vec{x} \times (\vec{a} + \vec{b}) = 0$

$\vec{x} = \mu(\vec{a} + \vec{b}) = \mu(3\hat{i} + 2\hat{j} - \hat{k})$

$|\vec{x}| = \mu\sqrt{14}$

$\left(\frac{x}{|\vec{x}|}.\sqrt{14}\right) = \frac{3\hat{i} + 2\hat{j} - \hat{k}}{\sqrt{14}}$

(R) Let AP = 1



$\Rightarrow$ $\overrightarrow{OP} = \overrightarrow{OA} \pm \overrightarrow{AP} = \overrightarrow{OA} \pm 1.\frac{\overrightarrow{AB}}{|\overrightarrow{AB}|}$

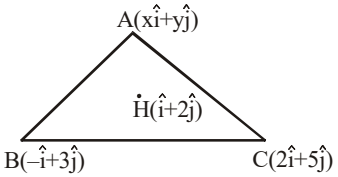
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$= \hat{i} - 2\hat{j} + \hat{k} \pm \frac{3\hat{j} - 4\hat{k}}{5}$

taking +ve sign $\frac{5\hat{i} - 7\hat{j} + \hat{k}}{5}$

(S) AH ⊥ BC $\Rightarrow \overrightarrow{AH}.\overrightarrow{BC} = 0$



$3x + 2y = 7$ .....(1)

AH ⊥ CA

$\Rightarrow 2x - y = 1$ .....(2)

$\Rightarrow x = \frac{5}{7}, y = \frac{17}{7}$

$\Rightarrow A = \frac{1}{7}(5\hat{i} + 17\hat{j})$

24. Ans. (A)

(P) Let $D(\vec{O}), A(\vec{a}), B(\vec{b}), C(\vec{c})$

given volume of tetrahedron is $\left|\frac{1}{6}[\vec{a}\ \vec{b}\ \vec{c}]\right| = \frac{81}{2}$

$|(\vec{a}\ \vec{b}\ \vec{c})| = 3^5$

Centroids by faces are

$G_1\left(\frac{\vec{a}+\vec{b}+\vec{c}}{3}\right), G_2\left(\frac{\vec{a}+\vec{b}}{3}\right), G_3\left(\frac{\vec{b}+\vec{c}}{3}\right), G_4\left(\frac{\vec{c}+\vec{a}}{3}\right)$

Volume of parallelopiped

$= \left|[\overrightarrow{G_1G_2}\ \overrightarrow{G_1G_3}\ \overrightarrow{G_1G_4}]\right|$

$= \left|\left[-\frac{\vec{c}}{3}\ -\frac{\vec{a}}{3}\ -\frac{\vec{b}}{3}\right]\right| = \left|\frac{[\vec{a}\ \vec{b}\ \vec{c}]}{3^3}\right| = 9$

(Q) Let the image of the point (1, 0, 1) in the plane x – y – z = 1 is $A'(a,b,c)$

D.R. of AA' are (a – 1, b, c – 1)

D. R. of normal are (1, –1, –1)

$\frac{a-1}{1} = \frac{b}{-1} = \frac{c-1}{-1} = \lambda \Rightarrow a = \lambda + 1, b = -\lambda, c = 1 - \lambda$

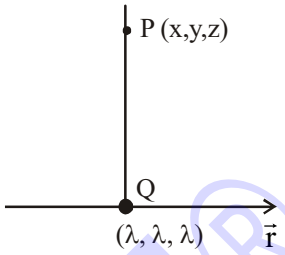
mid point of AA' lie on plane

$x - y - z = 1 \Rightarrow \lambda = \frac{2}{3}$

$a = \frac{5}{3}, b = \frac{-2}{3}, c = \frac{1}{3} \Rightarrow 3(a-b+c) = 3\left(\frac{8}{3}\right) = 8$

(R) $\vec{QP}.(i+j+k) = 0 \Rightarrow (x-\lambda)+(y-\lambda)+(z-\lambda) = 0$

$\lambda = \frac{x+y+z}{3}$



$\vec{r} = \lambda\ (i+j+k)$

$(PQ)^2 = 9 \Rightarrow (x-\lambda)^2 + (y-\lambda)^2 + (z-\lambda)^2 = 9$

$x^2 + y^2 + z^2 + 3\lambda^2 - 2\lambda(x+y+z) = 9$

$x^2 + y^2 + z^2 + 3\lambda^2 - 2\lambda.\ 3\lambda = 9$

$\Rightarrow x^2 + y^2 + z^2 - 3\left(\frac{x+y+z}{3}\right)^2 = 9$

$\Rightarrow x^2 + y^2 + z^2 - xy - yz - zx = \frac{27}{2}$

$k = 27, \quad \frac{k}{9} = 3$

(S) $\vec{C} = \lambda(\vec{a} \times \vec{b})$ also

$\left|\frac{1}{6}[\vec{a}\ \vec{b}\ \vec{c}]\right| = 1 \Rightarrow |[\vec{a}\ \vec{b}\ \vec{c}]| = 6$

$\Rightarrow |(\vec{a} \times \vec{b}).\vec{c}| = 6 \Rightarrow \frac{|c|^2}{|\lambda|} = 6 \Rightarrow 6 = 6|\lambda| \Rightarrow |\lambda| = 1$

$\therefore |c|^2 = \lambda^2 |a|^2 |b|^2 \sin^2\theta \Rightarrow 6 = 12\sin^2\theta = \theta = \frac{3\pi}{4} (a.b < 0)$

p + q = 7

SECTION–II (iii)

Numerical Grid Type (Upto Second Decimal place) (4 Marks each, –1 for wrong answer)

1. Ans. 1.25

$\vec{r}.(\vec{a}+\vec{b}+\vec{c}) = 0$

$\Rightarrow (\cos x + \cos y + 2)[\vec{a}\ \vec{b}\ \vec{c}] = 0$

$\Rightarrow \cos x + \cos y + 2 = 0$

$\Rightarrow \sin x = \cos y = -1$

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$\Rightarrow$ we take $x = -\frac{\pi}{2}$, $y = \pi$ so that $x^2 + y^2$ obtain its minimum value.

$\therefore$ Minimum value of $\frac{4}{\pi^2}\left(x^2 + y^2\right) = 5$

2. Ans. 1.00

Let $\vec{a} = x\hat{i} + y\hat{j} + z\hat{k}$

$\therefore\ x^2 + y^2 + z^2 = 1$ ...(1)

also $\vec{a} \times \left(\hat{i} + \hat{j} + \hat{k}\right) = \hat{i} - \hat{k}$

$\Rightarrow \hat{i}(y - z) - \hat{j}(x - z) + \hat{k}(x - y) = \hat{i} - \hat{k}$

comparing, we get

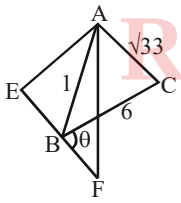
$z = x;\ y = 1 + x$

$\therefore$ By (1): $x = -\frac{2}{3}; y = \frac{1}{3}; z = -\frac{2}{3}$ or $x = z = 0, y = 1$

$\therefore\ \hat{a} = -\frac{2}{3}\hat{i} + \frac{1}{3}\hat{j} - \frac{2}{3}\hat{k}$

$\therefore\ \hat{a}.\hat{b} = -\frac{2}{3} + \frac{1}{3} - \frac{2}{3} = -\frac{3}{3} = -1$

3. Ans. 2.00



$\overrightarrow{AB}.\overrightarrow{AE} + \overrightarrow{AC}.\overrightarrow{AF} = 2$

$\overrightarrow{AB}.\left(\overrightarrow{AB} + \overrightarrow{BE}\right) + \overrightarrow{AC}.\left(\overrightarrow{AB} + \overrightarrow{BF}\right) = 2$

$\left|\overrightarrow{AB}\right|^2 + \overrightarrow{AB}.\overrightarrow{BE} + \overrightarrow{AC}.\overrightarrow{AB} + \overrightarrow{AC}.\overrightarrow{BF} = 2$ ...(1)

$\overrightarrow{AB} + \overrightarrow{CA} = -\overrightarrow{BC}$

$\left|\overrightarrow{AB} + \overrightarrow{CA}\right| = \left|-\overrightarrow{BC}\right|$

$\Rightarrow 1 + 33 - 2\overrightarrow{AB}.\overrightarrow{AC} = 36$

$\overrightarrow{AB}.\overrightarrow{AC} = -1$

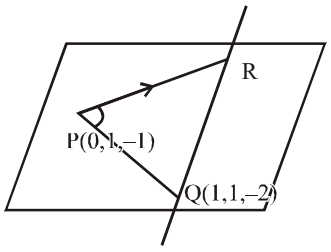
(1) $\Rightarrow \overrightarrow{AB}.\overrightarrow{BE} + \overrightarrow{AC}.\overrightarrow{BF} = 2$

$\left(-\overrightarrow{AB} + \overrightarrow{AC}\right).\overrightarrow{BF} = 2 \quad \left[\because \overrightarrow{BE} = -\overrightarrow{BF}\right]$

$\left(\overrightarrow{BA} + \overrightarrow{AC}\right).\overrightarrow{BF} = 2$

$\overrightarrow{BC}.\overrightarrow{BF} = 2$

4. Ans. 10.80



$\frac{x-1}{2} = \frac{y-1}{3} = \frac{z+2}{1} = \lambda$

$\therefore$ R$(2\lambda + 1, 3\lambda + 1, \lambda - 2)$

$\overrightarrow{PR}.\overrightarrow{PQ} = 0$

$((2\lambda + 1)\hat{i} + 3\lambda\hat{j} + (\lambda - 1)\hat{k}).(\hat{i} - \hat{k}) = 0$

$2\lambda + 1 - \lambda + 1 = 0 \quad \Rightarrow \quad \lambda = -2$

$\therefore$ R is $(-3, -5, -4)$

$PR = \sqrt{9 + 36 + 9} = \sqrt{54}$

5. Ans. 0.85 or 0.86

Let M being origin then one possible equation of face LNO will be $\Rightarrow x + \frac{y}{2} + \frac{z}{3} = 1$

$\therefore$ required altitude = distance of LNO from M $= \frac{6}{7}$

6. Ans. 4.50

equation of P_3 is $P_1 + \lambda P_2 = 0$

$\Rightarrow (x + y - 2z - 1) + \lambda(\alpha x - y + z - 2) = 0$

$\because$ P and Q lie on P_3

$\Rightarrow (\alpha - 6) + \lambda(\alpha^2) = 0$

and $\left(\frac{5}{3} - 2\alpha\right) + \lambda\left(\frac{5}{3}\alpha - 3 + \alpha\right) = 0$

$\Rightarrow 2\alpha^3 + \alpha^2 - 19\alpha + 18 = 0$

$\Rightarrow \alpha = 2$ is only rational value.

$\Rightarrow$ equation of planes are

$P_1 : x + y - 2z = 1$ and $P_2 : 2x - y + z = 2$

vector along L is

$$\vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -2 \\ 2 & -1 & 1 \end{vmatrix} = -\hat{i} - 5\hat{j} - 3\hat{k}$$

$\Rightarrow$ its direction cosines are

$\pm\left(\frac{1}{\sqrt{35}}, \frac{5}{\sqrt{35}}, \frac{3}{\sqrt{35}}\right)$

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